

GATE PE

Petroleum Engineering

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 20 questions (31%)

Moderate: 38 questions (58%)

Difficult: 7 questions (11%)

GENERAL INSTRUCTIONS

GATE PE — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

1M [-0.33]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q2.

1M [-0.33]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

- (A) Ce^x
- (B) Ce^{-x}
- (C) x^2
- (D) $\ln(x)$

Q3.

1M [-0.33]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q4.

1M [-0.33]

*[Reservoir Engineering]*Darcy's law: $q = ?$

- (A) $kA \Delta P / (4L)$
- (B) $;\ddot{A} \ddot{o}^\circ$
- (C) $kL/;\ddot{A}$
- (D) $;\hat{A} \hat{o}^\circ$

Q5.

1M [-0.33]

[Drilling Engineering]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q6.

1M [no -ve]

[Production Engineering]

ESP stands for:

Enter your answer here

Q7.

1M [no -ve]

[Petroleum Formation Evaluation]

Gamma ray log measures:

Enter your answer here

Q8.

1M [-0.33]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q9.

1M [-0.33]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q10.

1M [-0.33]

[Offshore Drilling]

Jack-up rig works in:

- (A) Deep water
- (B) Shallow water
- (C) Both
- (D) None

Q11.

1M [-0.33]

*[Reservoir Engineering]*Darcy's law: $q = ?$

- (A) $\frac{kA\Delta P}{\mu L}$
- (B) $\frac{\mu L}{kA\Delta P}$
- (C) $\frac{kL}{\mu \Delta P}$
- (D) $\frac{\mu \Delta P}{kL}$

Q12.

1M [no -ve]

[Production Engineering]

ESP stands for:

Enter your answer here

Q13.

1M [-0.33]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q14.

1M [no -ve]

[Offshore Drilling]

Jack-up rig works in:

*Enter your answer here***Q15.**

1M [no -ve]

[Petroleum Chemistry]

API gravity > 31.1 is:

*Enter your answer here***Q16.**

1M [no -ve]

[Production Engineering]

ESP stands for:

*Enter your answer here***Q17.**

1M [no -ve]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

*Enter your answer here***Q18.**

1M [-0.33]

[Drilling Engineering]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q19.

1M [no -ve]

[Production Engineering]

ESP stands for:

*Enter your answer here***Q20.**

1M [-0.33]

[Petroleum Formation Evaluation]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q21.

1M [-0.33]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q22.

1M [-0.33]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q23.

1M [-0.33]

[Drilling Engineering]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q24.

1M [-0.33]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q25.

1M [-0.33]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q26.

1M [-0.33]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q27.

1M [-0.33]

[Drilling Engineering]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q28.

1M [-0.33]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q29.

1M [-0.33]

[Production Engineering]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q30.

1M [no -ve]

[Petroleum Chemistry]

API gravity > 31.1 is:

*Enter your answer here***Q31.**

2M [no -ve]

*[Reservoir Engineering]*Darcy's law: $q = ?$

*Enter your answer here***Q32.**

2M [no -ve]

[Offshore Drilling]

Jack-up rig works in:

*Enter your answer here***Q33.**

2M [no -ve]

[Petroleum Exploration]

Source rock rich in:

*Enter your answer here***Q34.**

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q35.

2M [no -ve]

[Petroleum Exploration]

Source rock rich in:

*Enter your answer here***Q36.**

2M [-0.66]

[Production Engineering]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q37.

2M [-0.66]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q38.

2M [-0.66]

[Petroleum Chemistry]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q39.

2M [no -ve]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q40.

2M [no -ve]

[Petroleum Formation Evaluation]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q41.

2M [no -ve]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q42.

2M [-0.66]

[Petroleum Formation Evaluation]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q43.

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q44.

2M [-0.66]

[Petroleum Formation Evaluation]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q45.

2M [no -ve]

*[Reservoir Engineering]*Darcy's law: $q = ?$

*Enter your answer here***Q46.**

2M [no -ve]

[Petroleum Chemistry]

API gravity > 31.1 is:

*Enter your answer here***Q47.**

2M [no -ve]

*[Reservoir Engineering]*Darcy's law: $q = ?$

- (A) $kA\Delta P / \mu L$
- (B) $\rho g h / \mu$
- (C) kL / μ
- (D) $\rho g h$

Q48.

2M [no -ve]

[Petroleum Exploration]

Source rock rich in:

*Enter your answer here***Q49.**

2M [-0.66]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q50.

2M [no -ve]

[Petroleum Chemistry]

API gravity > 31.1 is:

*Enter your answer here***Q51.**

2M [no -ve]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q52.

2M [-0.66]

[Petroleum Formation Evaluation]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q53.

2M [-0.66]

*[Engineering Mathematics]*First order ODE $dy/dx = y$ solution:

- (A) Ce^x
- (B) Ce^{-x}
- (C) x^2
- (D) $\ln(x)$

Q54.

2M [no -ve]

[Production Engineering]

ESP stands for:

Enter your answer here

Q55.

2M [-0.66]

[Well Testing]

Pressure transient analysis uses:

- (A) Horner plot
- (B) Material balance
- (C) MBAL
- (D) PVT

Q56.

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q57.

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q58.

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q59.

2M [-0.66]

[General Aptitude]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Q60.

2M [-0.66]

[Drilling Engineering]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q61.

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q62.

2M [-0.66]

[Drilling Engineering]

Roller cone bit has:

- (A) 1 cone
- (B) 2
- (C) 3
- (D) 4

Q63.

2M [no -ve]

[Petroleum Formation Evaluation]

Gamma ray log measures:

*Enter your answer here***Q64.**

2M [no -ve]

[Production Engineering]

ESP stands for:

*Enter your answer here***Q65.**

2M [-0.66]

[Petroleum Exploration]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale